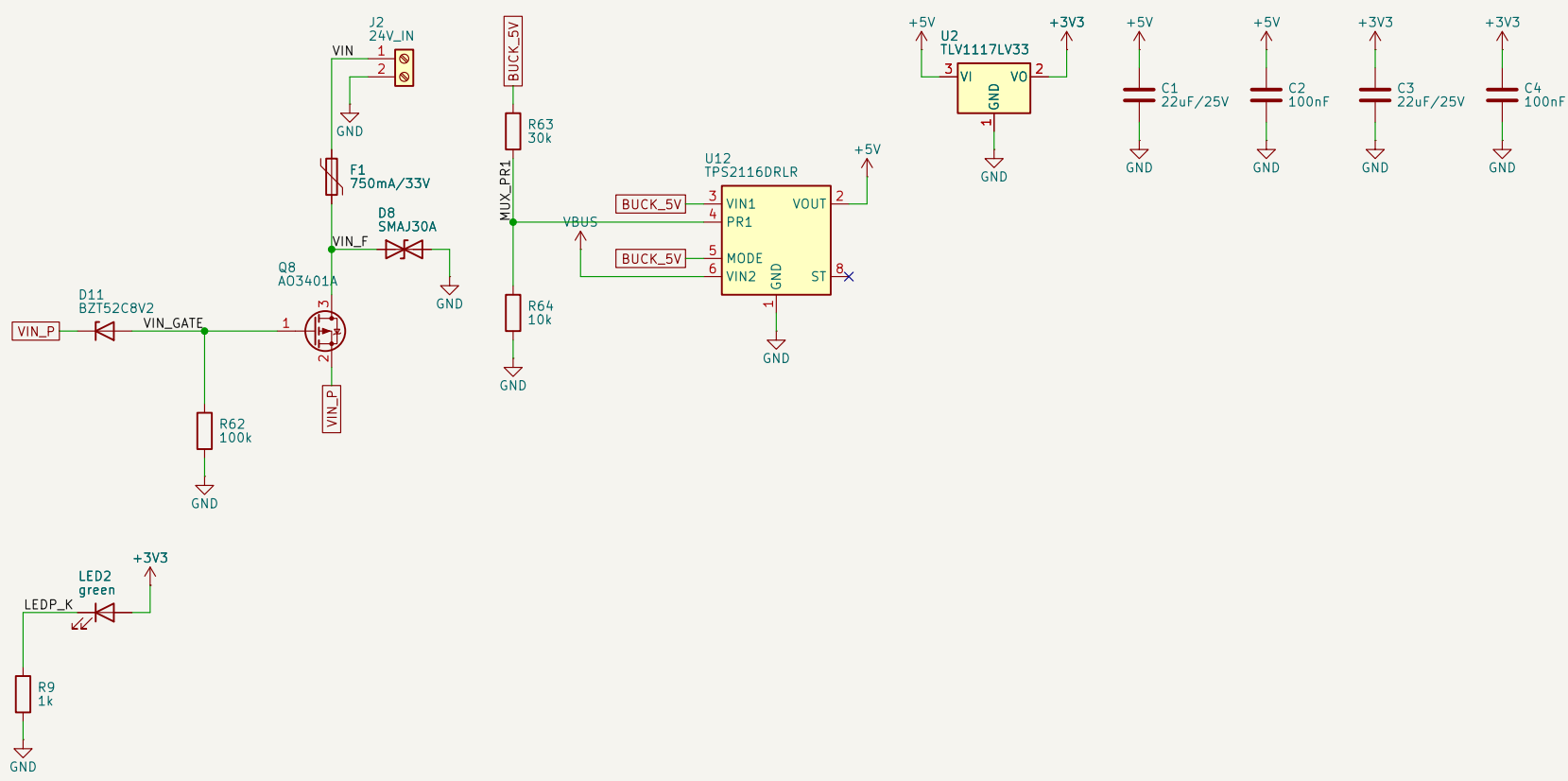
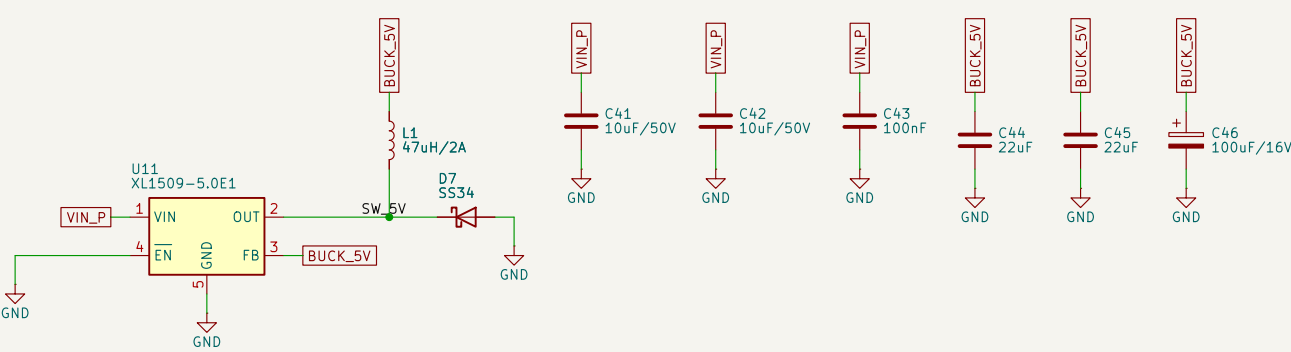


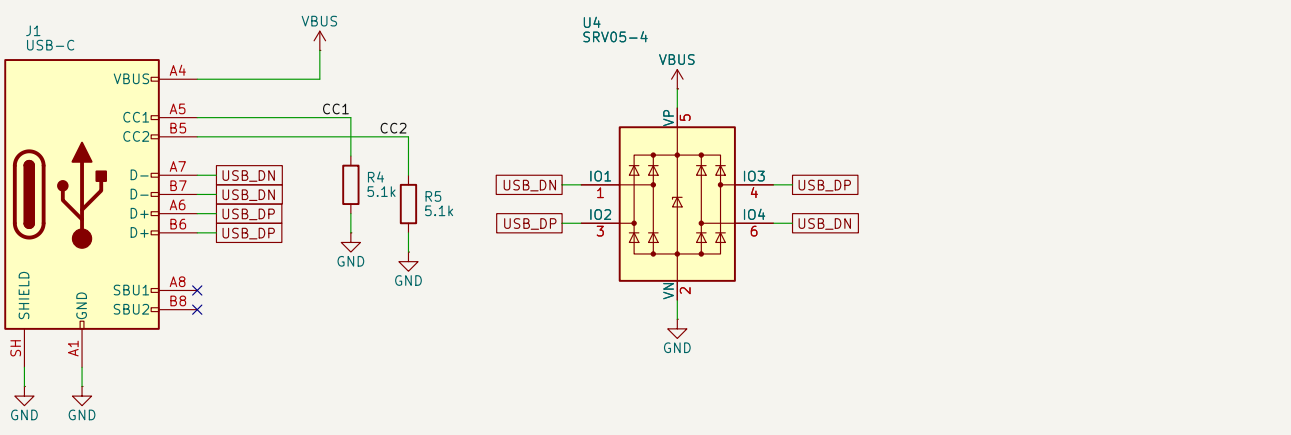
POWER IN 24 VDC terminal or USB  
F1+D8 clamp-and-blow at the entry. Q8 blocks reverse polarity as a P-FET, not a diode: its body diode conducts on correct polarity and the gate then pulls itself on through R62, clamped by D11 because 24 V is twice the A03401A's V<sub>GS</sub> rating. U11 (right) makes BUCK\_5V; U12 selects between it and USB VBUS to make +5V, giving the buck PRIORITY via R63/R64 rather than letting the higher voltage win -- a 5.25 V host would otherwise source the SSR terminals and the relay bank. U12 also blocks USB back-feeding the unpowered buck. U2 drops +5V to +3V3



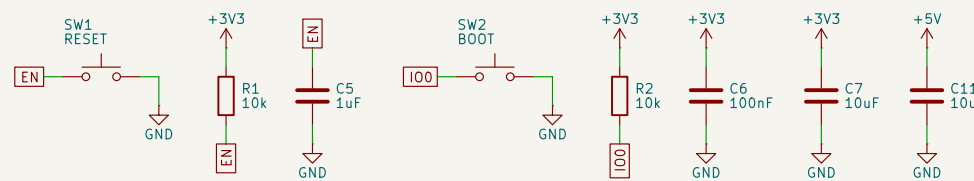
24 V -> 5 V BUCK XL1509-5.0, fixed output  
C41-C43 in, D7 catch, L1 47uH, C44/C45/C46 out. F8 ties to the output because the part is the fixed 5.0 V version; EN is active LOW and grounded. C46 is an ELECTROLYTIC on purpose: voltage-mode with fixed compensation needs its ESR zero, and an all-ceramic output has none. Keep C41/C42 - U11 - D7 tight: that loop, not L1, is what radiates.



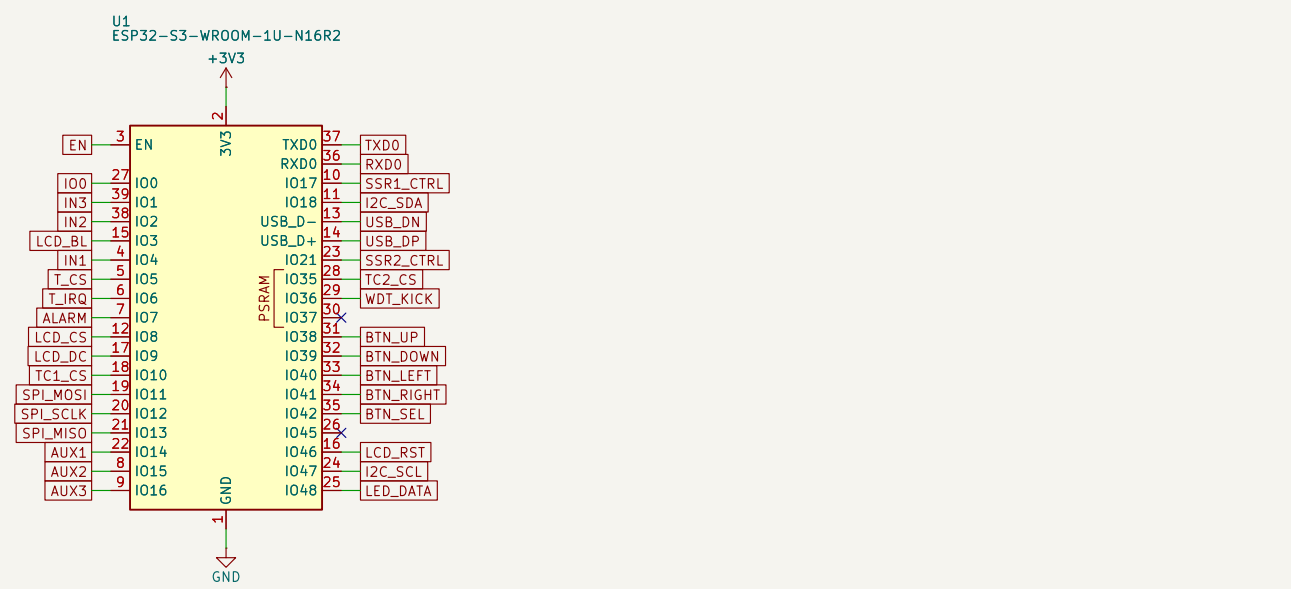
USB-C  
native USB flashing + ESD



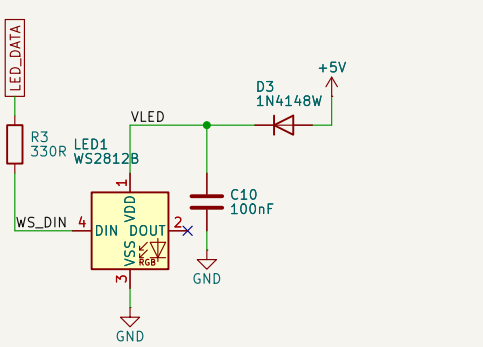
RESET / BOOT / DECOUPLING



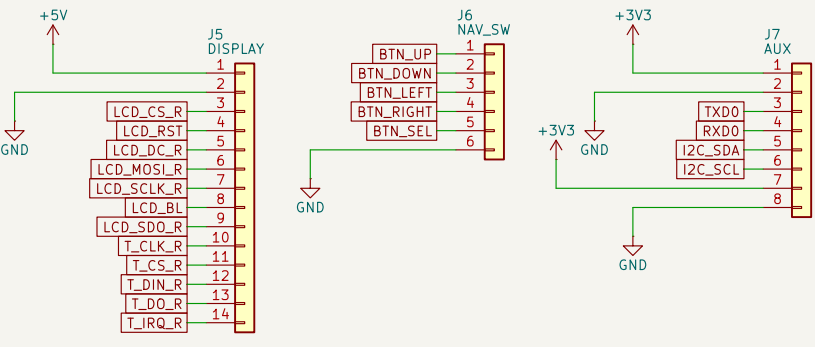
ESP32-S3-WROOM-1U-N16R2  
GPIOs = firmware Kconfig defaults



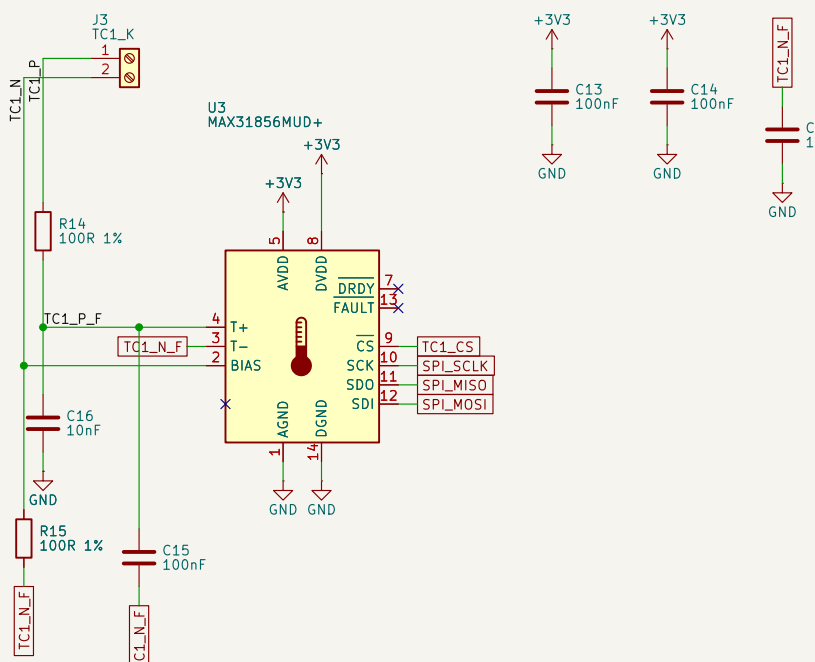
WS2812B STATUS LED  
VDD dropped -4.6V for 3.3V data margin



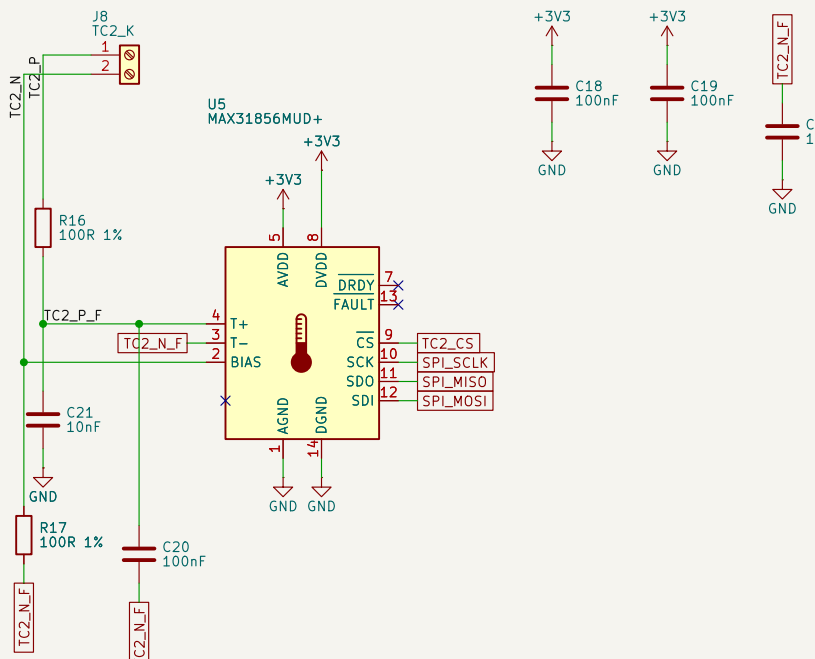
HEADERS  
J5 display+touch (14-pin), J6 nav, J7 aux+I2C



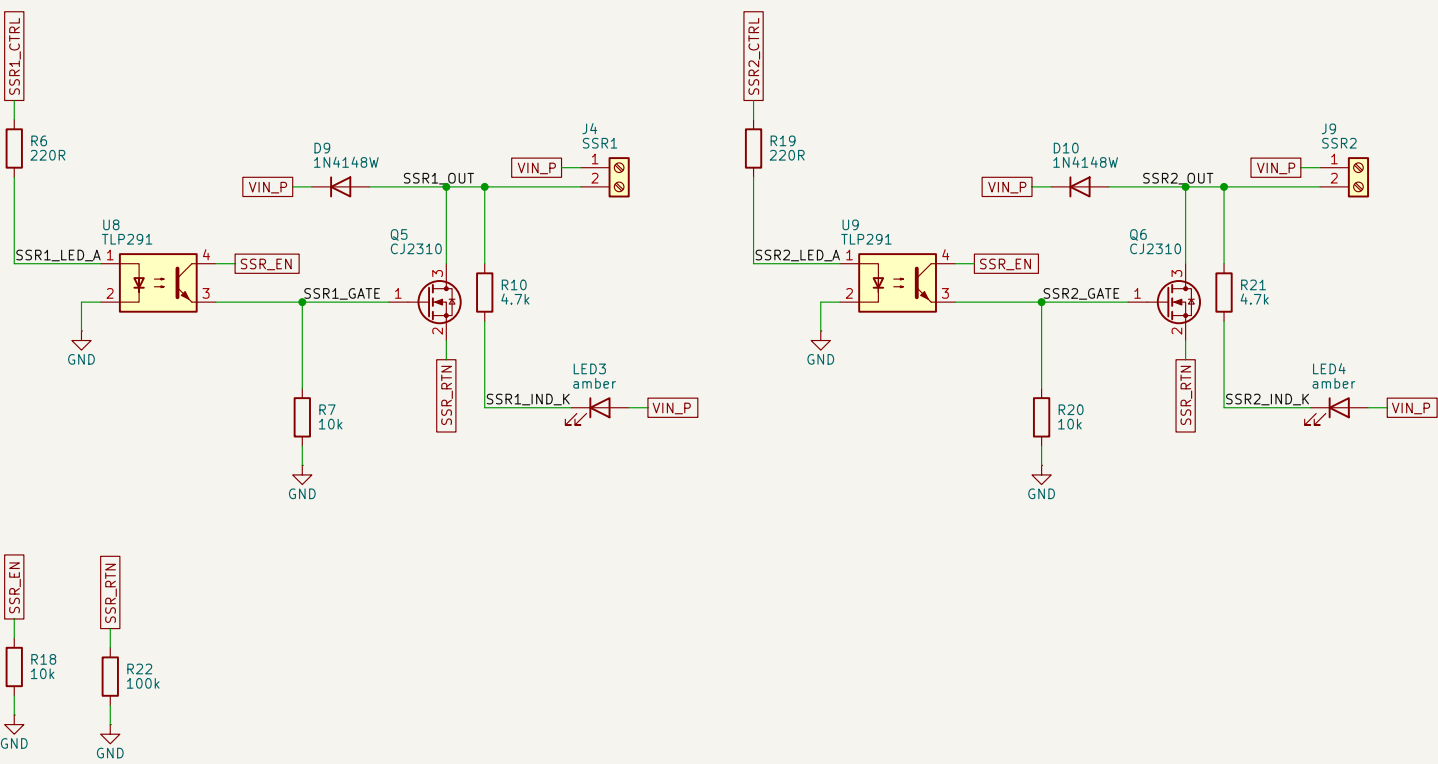
THERMOCOUPLE 1 (control) MAX31856  
T- floats to J3, biased via BIAS



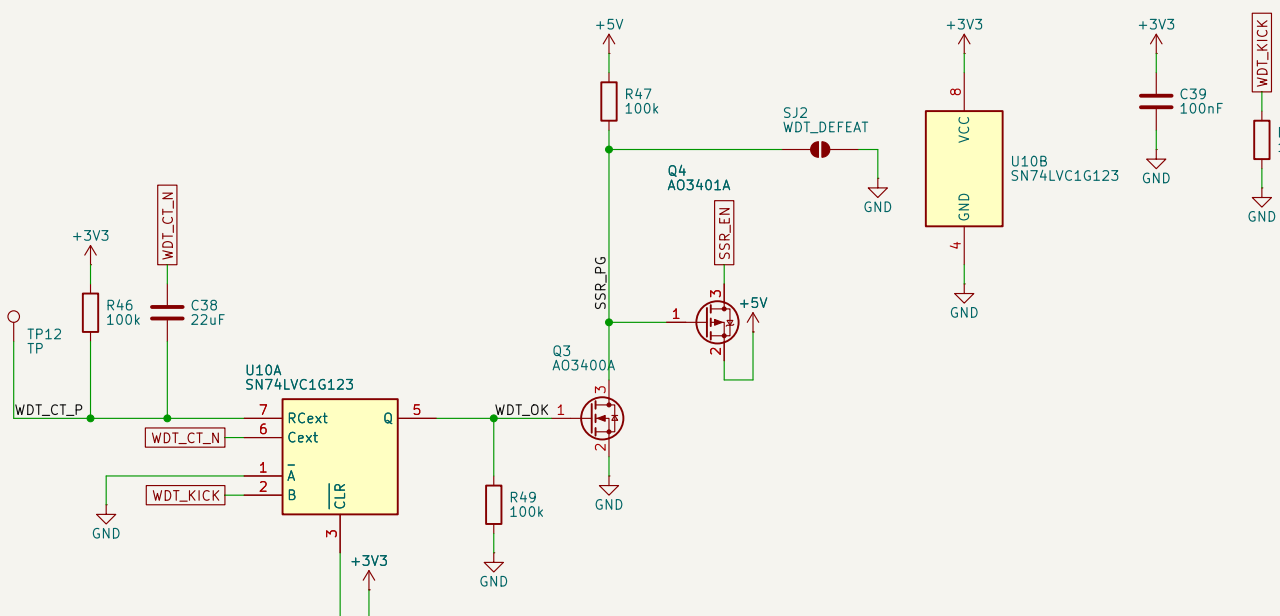
THERMOCOUPLE 2 (load) MAX31856  
T- floats to J8, biased via BIAS



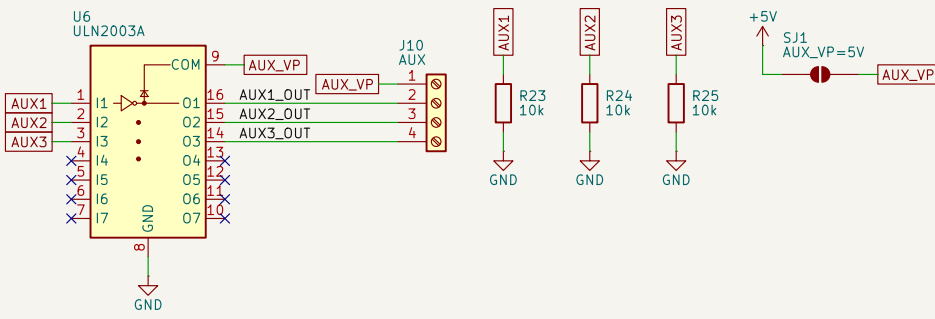
SSR DRIVE x2  
24 V terminals off VIN\_P, switched low side by a 60 V CJ2310. The GPIO drives an opto LED, never the gate: at 24 V a gate-to-drain failure would otherwise put 240 mA into the pin. Opto collectors sit on SSR\_EN (5 V), so the gate lands at -4.9 V from the RAIL rather than from CTR. Q7 is the watchdog's gate in the shared return - it opens a channel whose MOSFET failed short. R18/R22 define SSR\_EN and SSR\_RTN when their switches are off. NOT an isolation barrier - see design.py's SSR block.



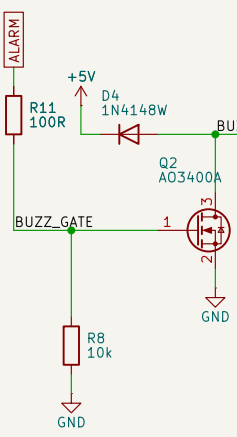
HARDWARE WATCHDOG  
U10 retriggerable one-shot: every rising edge on WDT\_KICK restarts a -2.2 s window, so Q3 stays high only while firmware keeps kicking. Q holds Q3 on; Q3 pulls SSR\_PG down, turning on high-side Q4, which supplies SSR\_EN - the +5V GATE DRIVE rail feeding both opto collectors and Q7's gate. R46/C38 set the window. R48 holds B low when the MCU is high-Z. R47 is the fail-safe pull-up. SJ2 = bring-up defeat, REMOVE for service.



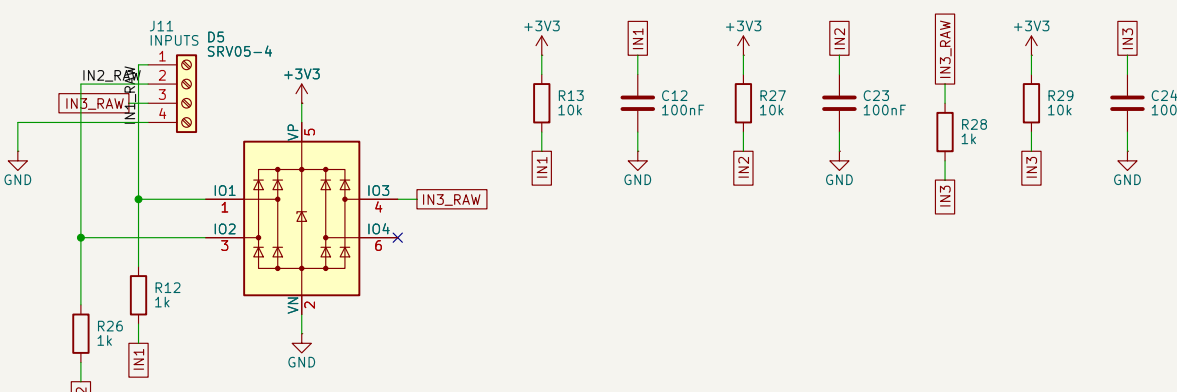
AUX OUTPUT BANK  
ULN2003A vent/purge/spare; COM->AUX\_VP, SJ1 links to +5V



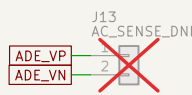
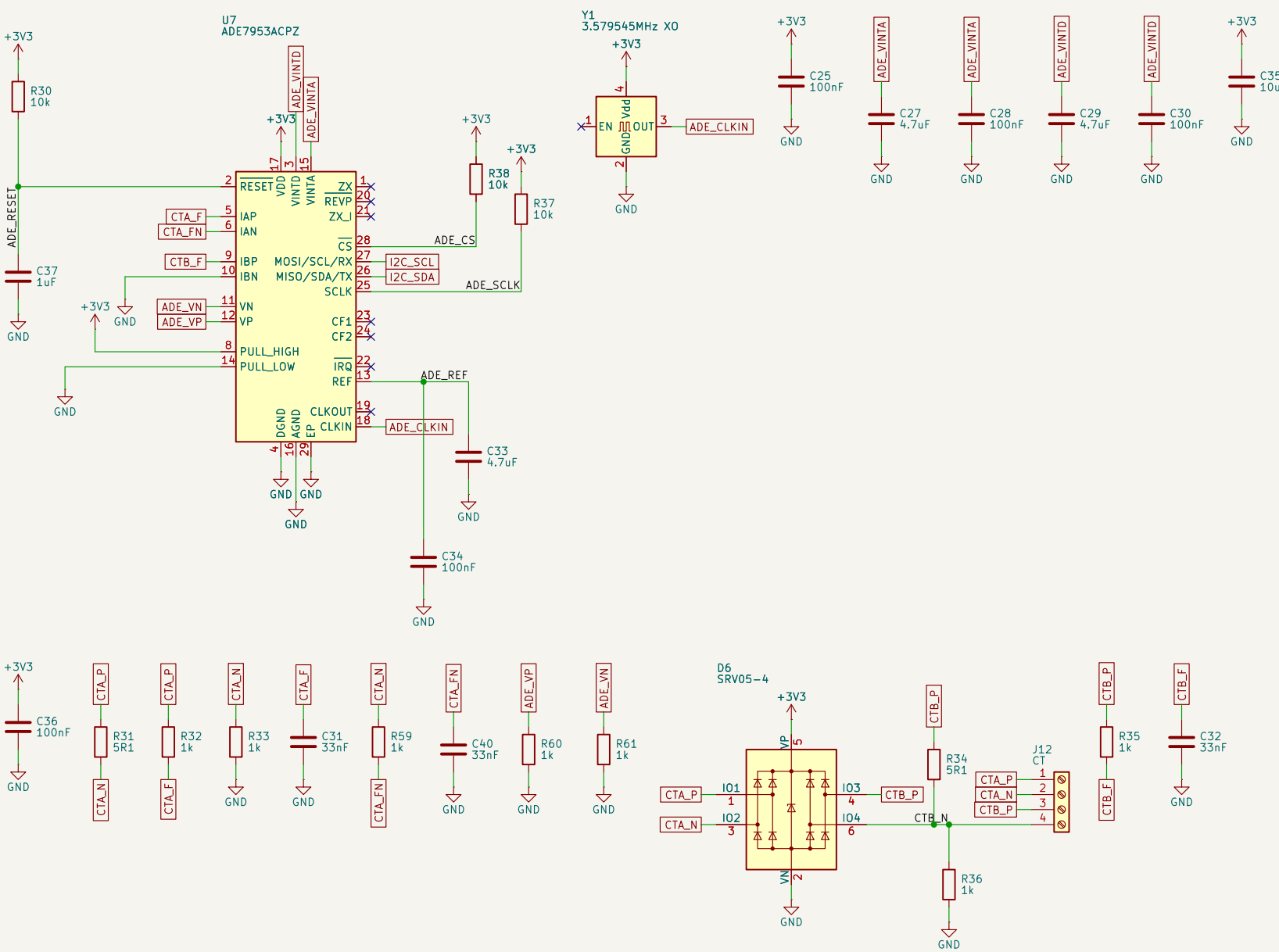
ALARM BUZZER



PROTECTED INPUTS x3 I<sub>ld</sub> / gas-flow / spare  
1k + 10k pull-up + 100nF each, SRV05-4 TVS



CT CURRENT SENSING ADE7953, I2C, current-only  
no mains - VP/VN to DNP J13



NOTES - Bisque kiln controller, rev B

SSR terminals J4/J9: pin1 = VIN\_P (24 V, fused and TVS-clamped, ALWAYS LIVE), pin2 = the switched low side. The watchdog gates the shared return (Q7), not the supply, so a tripped watchdog leaves 24 V on pin 1 and opens pin 2. Boot-safe: R7/R20 hold each gate down while the optos are dark. The optos are NOT an isolation barrier - both sides share GND. Rev B's opto-ISOLATED design was reverted and stays reverted: U8/U19 are here to keep a 24 V gate fault off the GPIO. Per channel <=200 mA: F1 holds at 750 mA and U11 takes -250.

Nav switch J6 is panel-mounted: inputs use ESP32 pull-ups.

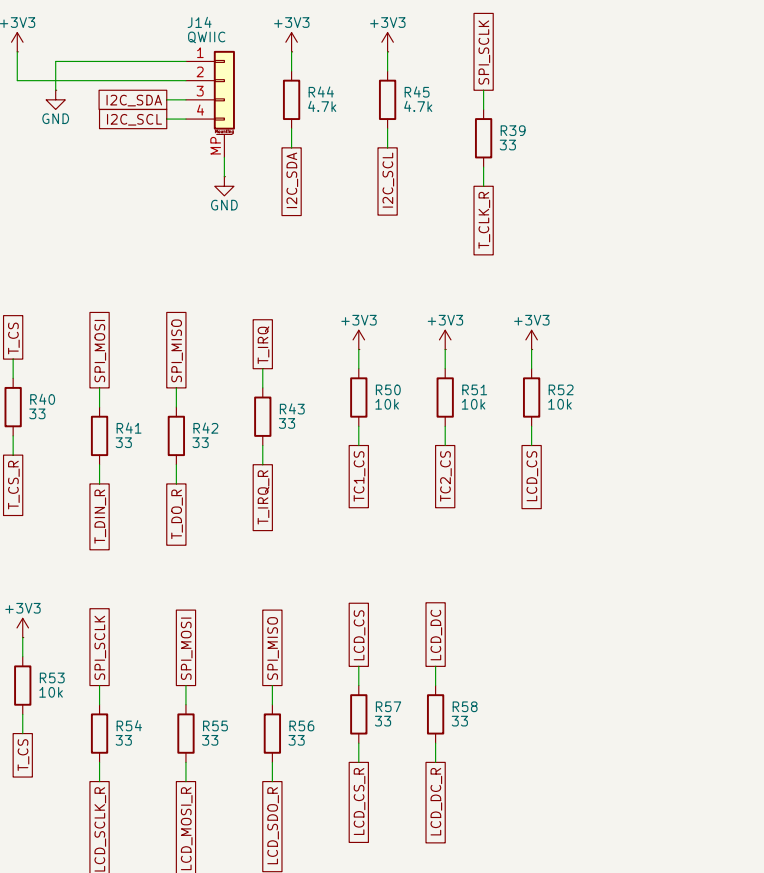
J11: IN1/IN2/IN3 (I<sub>ld</sub>/gas-flow/spare) + GND - dry contact each channel to GND.

Display J5 (14-pin, ST7796S + XPT2046 touch module, LCDWIKI MSP4021): 1+=5V 2=GND 3=CS 4=RTST 5=DC 6=SD0/MOSI 7=SSCK 8=BL 9=SDV/MISO 10=LT CLK 11=+CS 12=L DIN 13=L DIN 14=L RD. PIN 1 IS +5V NOT 3V3 - the module regulates on board: do not wire a 3.3V-only panel to it. Logic is 3.3V: touch shares SPI2 with the LCD and both MAX31856s (R39-R43 damping).

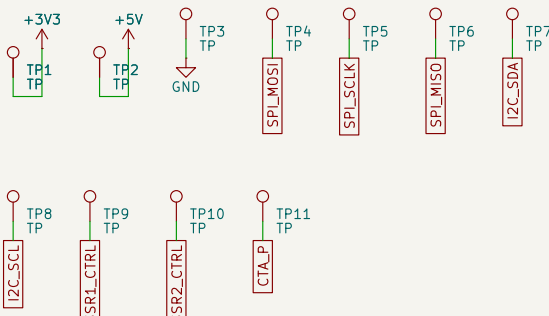
J12: CT current inputs, CTA\_P/CTA\_N/CTB\_P/CTB\_N - one CT clamp per SSR zone.

J13: DNP SELV voltage sense header, NOT mains-rated, not fitted - do not wire to AC mains. V2/C25/C26 load caps are an ASSUMED value (unverified C.L.), see design.py comment. ADE7953 I2C address 0x38 collides with PCT874A.

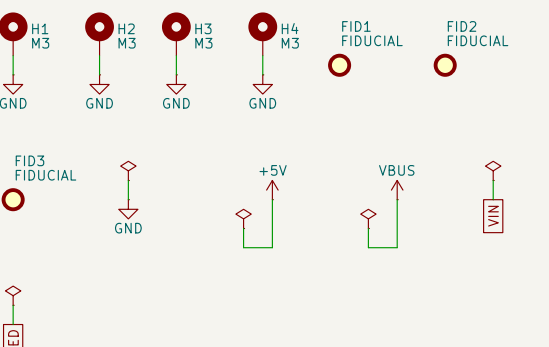
SPI BUS DAMPING + CHIP-SELECT PULL-UPS + I2C EXPANSION  
R39-R43 damp the display module's XPT2046 touch lines and R54-R58 the four 40 MHz display lines; both loom to J5. R56 is also the SDO hedge - lift it if the panel does not tri-state J5.9 off the shared SPLMISO. R50-R53 hold all four chip selects deasserted from power-on, before firmware configures any GPIO. J14 Qwic and J7 5-8 (0.1 in) share I2C, pulled up by R44/R45



TEST POINTS



MOUNTING / FIDUCIALS / POWER FLAGS  
FID1-3 are bare-copper pick-and-place targets, on no net



4-layer, 100 x 100 mm, JLPCB standard process  
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953

Sheet: 7  
File: bisque-controller.kicad\_sch

Title: Bisque Kiln Controller

Size: A1  
Kicad E.D.A. 10.0.6

Rev: B.2  
1 of 1